

Hannah Bowley

DATA SCIENTIST

Miami, FL

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Data science and research analytics leader with expertise in higher education analytics, statistical modeling, machine learning, and applied AI, translating complex data into actionable insights that support student success, research compliance, and institutional decision-making.

Professional Experience

Director, Research and Data Analytics

Miami, FL

MIAMI DADE COLLEGE, DIVISION OF STRATEGY AND INSTITUTIONAL EFFECTIVENESS

November 2023 - Present

- Lead the Institutional Research and Data Analytics team in delivering institution-wide analytics and reporting that support strategic planning, executive decision-making, accreditation, and state and federal compliance.
- Managed and mentored a team of data professionals, strengthening technical capacity in statistical modeling, machine learning, SQL, object-oriented programming in R and Python, data visualization using Power BI and Shiny, and reproducible research workflows using Quarto.
- Led the development of machine learning models to predict course success outcomes and student retention, enabling earlier identification of at-risk students and informing targeted academic interventions.
- Designed and prototyped agentic AI systems to automate and triage Kaseya ticket requests, streamlining Institutional Research data requests and reducing manual intake and turnaround time.
- Served as Chair of the Institutional Review Board (IRB) and Co-Chair of the CASSC Research and Testing Committee, providing oversight for research compliance, ethical review, and institutional assessment practices.
- Advised executive leadership on enterprise AI and cloud solutions to integrate siloed data systems and expand analytic capacity collegewide.

Principal Data Scientist

Miami, FL

PORTABLE NETWORKS

August 2023 - Present

- Served in a client-facing consulting role, leading the design, development, and deployment of custom business intelligence solutions.
- Store Performance Tracking System | Abcaco Foods: Developed and maintain a Shiny application on AWS with a PostgreSQL backend, enabling real-time sales data entry for field teams, product placement tracking, and manager engagement, improving operational efficiency and sales.
- Business Intelligence Platform | Lucto Fire: Developed and Deployed a Shiny-based BI application integrating Salesforce Reports that provided real time tracking of active projects, forecasted workforce demand and allocation to ensure timely contractor hiring and resource planning.

Research Coordinator

Miami, FL

FLORIDA INTERNATIONAL UNIVERSITY- PROJECT ON LANGUAGE AND SPATIAL DEVELOPMENT

February 2020 - October 2023

- Coordinated the Project Seahorse Study, examining relationships between hippocampal structure and function and spatial ability.
- Led independent statistical modeling of trace eyeblink conditioning to assess individual and sex differences in learning and memory processes.
- Assisted with development of supervised machine learning models for spatial language analysis to identify patterns in language use.
- Managed longitudinal data analysis and dissemination of findings through conference presentations and peer-reviewed publications.
- Oversaw R01 grant operations, including IRB protocol development and amendments, progress reporting, and budget management.
- Directed participant recruitment, scheduling, and completion of behavioral, neuroimaging, and parent-child interview visits.
- Supervised and trained a team of research assistants and ensured quality control of neuroimaging outputs and study data entry.

Teaching Fellow

Remote

RESEARCH CENTER FOR MINORITY INSTITUTIONS (RCMI) CONSORTIUM

October 2022 - May 2023

- Data Science instructor of R programming Language using R for Reproducible Scientific Analysis Data Carpentry curriculum

Research Assistant and Biostatistics Practicum

Miami, FL

FLORIDA INTERNATIONAL UNIVERSITY, VARIOUS NIH FUNDED PROJECTS

August 2018 - December 2022

- Supported analytics for NIH-funded neurodevelopmental research, including the Adolescent Brain Cognitive Development (ABCD) Study.
- Generated and curated structured datasets from neuropsychological and clinical instruments, including Q-Interactive, NIH Toolbox, PPVT, TVIP, KSADS, and medication inventories.
- Performed data quality assurance, preprocessing, and validation for longitudinal behavioral, clinical, neuroimaging, and physiological datasets in preparation for analysis.
- Conducted neuroimaging data processing, including ROI tracing and quality control of structural and functional MRI data using Mango/AFNI.
- Designed and implemented R-based statistical workflows for missing data analysis, propensity score matching, and multilevel modeling to evaluate the effects of stimulant medication on diffusion-weighted neuroimaging metrics.
- Developed visualization and reporting scripts to assess data integrity and communicate analytic results.

Research Assistant

Tallahassee, FL

FLORIDA STATE UNIVERSITY- WILBER LAB

August 2017 - August 2018

- Conducted analysis of behavioral and neural data from rodent spatial navigation experiments using MATLAB for spike sorting/unsorting.
- Supported behavioral data collection for a project examining spatial sequence learning and its relation to the parietal cortex
- Developed technical documentation and visual materials to support data dissemination, research communication and grant submissions.

Education

Florida International University

Miami, FL

MASTER OF PUBLIC HEALTH- BIostatISTICS

2022

- Practicum: Effects of Stimulant Medication on Brain Development – Propensity Matching and multi-level modeling of medication status on diffusion-weighted neuroimaging metrics

Florida State University

Tallahassee, FL

BACHELOR OF SCIENCE- PSYCHOLOGY

2018

- Federal Work Study Project: Parietal contributions to context dependent spatial sequence learning

Certifications

Data Engineering	Data Engineering Professional Certificate – Snowflake
Generative AI	Generative AI Professional Certificate – Snowflake
IRB Member Training	CITI IRB Member Training
IRB Chair Training	CITI IRB Chair Training

Recent Projects

Real Time Disaster Response Analytics Platform

Miami, FL

CATALYST MIAMI/CEFA-CLEAN ENERGY FOR ALL

October 2024

- Built a Python and R-based automated pipeline using Selenium and Google Sheets API to track utility outages, extreme heat, and boil water notices during Hurricane Milton. Centralized real-time data used to activate community response teams, enhancing disaster relief coordination and scalability for future emergency response deployments.

Environmental Health Scan

Miami, FL

UNIVERSITY OF MIAMI, SYLVESTER COMPREHENSIVE CANCER CENTER

September 2023 - November 2023

- Conducted an environmental health analysis for the Florida Department of Health, leveraging R and automated data extraction to assess lung cancer incidence trends. Developed a data-driven community mitigation plan that informed tobacco cessation efforts across seven Florida counties.

Talks

MDC Boom! Letting Data Driven Strategic Interventions for Student Success

March 2024

HEUG ALLIANCE

Integrating R & Quarto into Institutional Research Data Workflows

February 2024

FLORIDA ASSOCIATION FOR INSTITUTIONAL RESEARCH

Interpreting Data for Public Health Research, Policy or Practice

June 2023

ROBERT STEMPEL COLLEGE OF PUBLIC HEALTH AND SOCIAL WORK, HEALTH PROMOTION AND DISEASE PREVENTION

Using R for day-to-day Research Workflows

April 2023

ROBERT STEMPEL COLLEGE OF PUBLIC HEALTH AND SOCIAL WORK, HEALTH PROMOTION AND DISEASE PREVENTION

Publications

1. Hall, L., Rengel, M., Overton, D., Alvarez-Vargas, D., Abad, C., Bowley, H., & Pruden, S. (2026). Parent mental rotation ability predicts less negation talk with children during 3D construction but not 2D drawing tasks. *Cognitive Development*, 79, 101750. <https://doi.org/https://doi.org/10.1016/j.cogdev.2026.101750>
2. Mattox, N., Bowley, H., Vieites, V., Wu, Y., Ralph, Y., Lioi, P., Rodriguez, V., Olazabal, K. S., Rengel, M., Hayes, T., Dick, A. S., Mattfeld, A., & Pruden, S. M. (2026). The relation between spatial language comprehension and mental transformation during early childhood. *Cognitive Development*, 77, 101663. <https://doi.org/https://doi.org/10.1016/j.cogdev.2025.101663>
3. Hall, L. V., Rengel, M., Bowley, H., Alvarez-Vargas, D., Abad, C., Overton, D., & Pruden, S. M. (2023). “You did a great job building that!” Links between parent-child prosocial talk and spatial language. *Developmental Psychology*, 59(9), 1676.

Presentations

1. Wu, Y, Bowley, H, Lioi, P, Ralph, Y, Vieites, V, Rengel, M, Renfro, A, Hayes, T, Dick, A.S, Mattfeld, A, & Pruden, S.M. (2024). *Wayfinding experience is related to spatial skills in 4- to 6-year-olds.*
2. Bowley, H, Wu, Y, Lioi, P, Vieites, V, Ralph, Y, Rengel, M, Renfro, A, Hayes, T, Dick, A.S, Mattfeld, A, & Pruden, S.M. (2023). *Exploring Individual and Sex Differences in Trace Eyeblink Conditioning (EBC) in 4- to 6- year olds.*
3. Wu, Y, Bowley, H, Lioi, P, Ralph, Y, Vieites, V, Rengel, M, Renfro, A, Hayes, T, Dick, A.S, Mattfeld, A, & Pruden, S.M. (2023). *The Association between Working Memory, Inhibitory Control, and Spatial Ability in 4- to 6- year-olds.*
4. Hall, L, Rengel, M, Bowley, H, Alvarez-Vargas, D, Abad, C, Overton, D, & Pruden, S.M. (2022). *Quantity and quality of parent-child spatial talk: The roles of prosocial talk and negative talk.*
5. Hall, L, Rengel, M, Bowley, H, Overton, D, Alvarez-Vargas, D, Abad, C, & Pruden, S.M. (2022). *Prosocial Talk During Collaborative Block Building Predicts Quantity of Spatial Language Production.*
6. Overton, D, Hall, L, Rengel, M, Bowley, H, Alvarez-Vargas, D, Abad, C, & Pruden, S.M. (2022). *Parent and child math language and relations to child spatial ability.*
7. Garcia, N, Rengel, M, Bowley, H, & Pruden, S.M. (2021). *Factors that Influence Mother- Child Spatial Play.*
8. Dick, A.S, Vieites, V, Renfro, A, Bowley, H, Rengel, M, Hayes, T, Mattfeld, A, Reeb-Sutherland, B, & Pruden, S.M. (2020). *Relation between hippocampal neurite density and trace eyeblink conditioning in four-to-six-year-old children.*
9. Hall, L, Rengel, M, Bowley, H, Hernandez, D, Giles, L, & Pruden, S.M. (2020). *A Novel Approach for Spatial Language Analysis: A supervised Machine Learning Algorithm.*
10. Bruce, N, Simmons, C.M, Bowley, H, Kirlan, M, Crafton, B, Pei, E, Baiamonte, M, & Wilber, A. A. (2018). *Inactivation of the Parietal Cortex and its effect on sequence learning.*
11. Simmons, C.M, Bowley, H, Kirlan, M, Baiamonte, M, & Wilber, A. A. (2018). *Pariteal contributions to context dependent spatial sequence learning.*

Skills

Programming & Statistical Tools	R, Python, SQL, SAS
Cloud & Data Platforms	AWS, Snowflake, PostgreSQL, SQL Server, DuckDB
Machine Learning & AI	Statistical Modeling, Machine Learning, Generative AI, Time Series Forecasting, Feature Engineering
Data Engineering & Automation	ETL Pipelines, API Integration, Task Scheduling, Bash/Shell Scripting
Visualization & Reporting	Shiny, Power BI, Quarto, R Markdown
DevOps & Version Control	Git, GitHub, GitHub Actions, Docker
Research Systems	REDCap, Qualtrics
Languages	English, Spanish